Why complex numbers arise

Real rotation matrix

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

has no nonzero real eigenvector when $\theta \not\equiv 0 \pmod{\pi}$. Its eigenvalues are

$$\lambda = e^{\pm i\theta}.$$

Complex eigenvectors describe oscillations, roots of unity, and periodic signals. A real matrix has conjugate-paired nonreal eigenvalues. Nonreal modes must be paired with conjugate coefficients to construct real solutions; real eigenvector modes can appear on their own.

We use $i^2 = -1$. Electrical engineering often writes j for the same imaginary unit.

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Arithmetic in the complex field

Write $z = a + bi$ with real a, b. Add and subtract components.

$$(a + bi) + (c + di) = (a + c) + (b + d)i.$$

Multiplication gives

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

The complex numbers form a field. For nonzero $a + bi$,

$$\frac{1}{a + bi} = \frac{a - bi}{a^2 + b^2}.$$

Multiplication commutes, and the positive squared modulus of a nonzero number supplies the reciprocal denominator. Standard basis vectors also work in $\mathbb{C}^n$, with complex coefficients.

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Conjugation, modulus, and dimension

Conjugate:

$$\overline{a + bi} = a - bi.$$

Modulus:

$$|z| = \sqrt{z\overline{z}} = \sqrt{a^2 + b^2}.$$

Conjugation preserves sums and products.

$$\overline{z + w} = \overline{z} + \overline{w}, \qquad \overline{zw} = \overline{z}\,\overline{w}.$$

Also $|zw| = |z||w|$, and the triangle inequality holds.

The complex dimension of $\mathbb{C}^n$ is n. Its real dimension is $2n$, because every component has a real and imaginary part. Specify the scalar field when counting dimensions or ranks.

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Polar form and Euler's formula

A nonzero complex number

$$z = a + bi$$

has

$$r = |z|, \qquad \theta = \operatorname{atan2}(b, a),$$
$$z = re^{i\theta} = r(\cos\theta + i\sin\theta).$$

Euler formula:

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Multiplication multiplies moduli and adds arguments.

$$(re^{i\theta})(se^{i\phi}) = rse^{i(\theta+\phi)}.$$

Powers:

$$(re^{i\theta})^k = r^k e^{ik\theta}.$$

For example,

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Polar form and Euler's formula (continued)

$$1 + i = \sqrt{2} e^{i\pi/4},$$
$$(1 + i)^8 = 16 e^{2\pi i} = 16.$$

Arguments differing by 2π represent the same number. A branch choice matters for multivalued operations such as the complex logarithm.

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Roots of unity

Solve

$$z^n = 1.$$

Set

$$\omega = e^{2\pi i/n}.$$

The roots are

$$1, \omega, \omega^2, \ldots, \omega^{n-1}.$$

They are equally spaced on the unit circle, forming a regular polygon for $n \geq 3$.

If $z = re^{i\theta}$ and $z^n = 1$, then

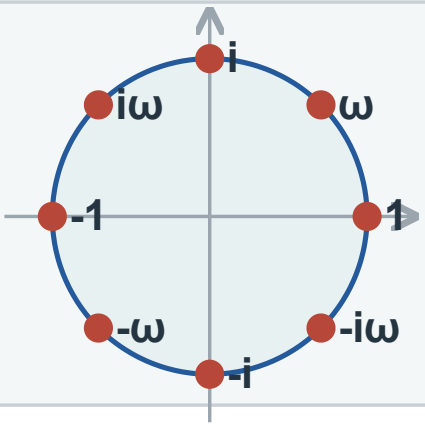
$$r = 1, \qquad n\theta = 2\pi k.$$

For $n = 4$, the roots are $1, i, -1, -i$. They also generate Fourier vectors and eigenvectors of cyclic shifts.

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Roots of unity (continued)

Eighth roots of unity: equally spaced on the unit circle



$\omega = e^{(2\pi i/8)}$

roots：$1, \omega, \ldots, \omega^7$

$F_8 F_8^H = 8I$

Orthogonal columns; Fourier basis

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The complex inner product

Our convention is

$$\langle u, v \rangle = u^H v = \sum_j \overline{u_j} v_j,$$

where

$$u^H = \overline{u^T}.$$

For $u = (1, i)^T$,

$$\|u\|^2 = u^H u = |1|^2 + |i|^2 = 2,$$

rather than the bilinear expression $1^2 + i^2 = 0$.

Inner product properties:

$$\langle u, v \rangle = \overline{\langle v, u \rangle},$$
$$\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle,$$
$$\langle u, \alpha v \rangle = \alpha \langle u, v \rangle,$$

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The complex inner product (continued)

$$\langle \alpha u, v \rangle = \overline{\alpha} \langle u, v \rangle.$$

Conjugation belongs to the first argument throughout. A different convention requires consistent changes in every formula.

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The adjoint moves a map across an inner product

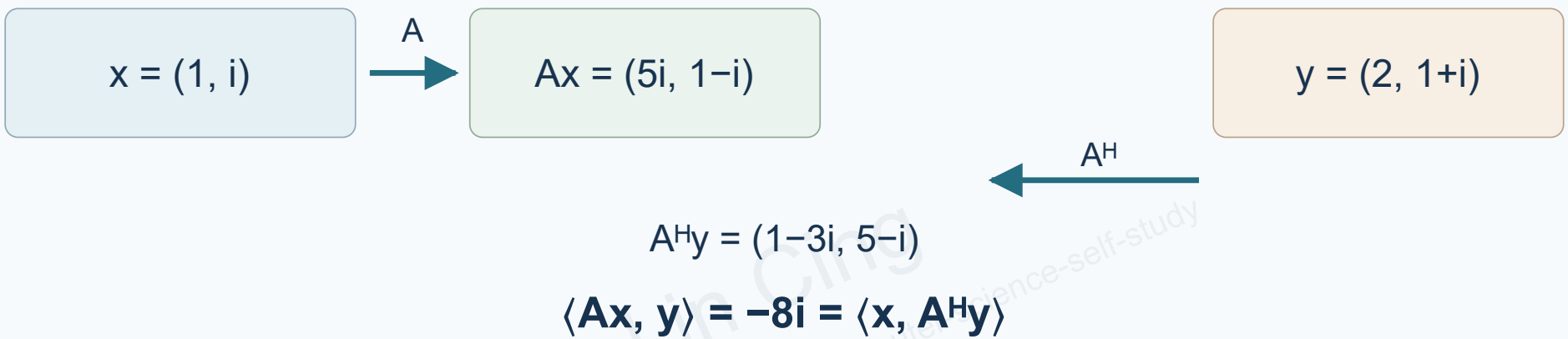
With this chapter's convention $\langle x, y\rangle = x^H y$, the adjoint of $A \in \mathbb{C}^{m\times n}$ is $A^H \in \mathbb{C}^{n\times m}$, since for all $x \in \mathbb{C}^n$, $y \in \mathbb{C}^m$,

$$\langle Ax, y\rangle = (Ax)^H y = x^H A^H y = \langle x, A^H y\rangle.$$

These are standard orthonormal coordinates: a plain transpose omits the necessary conjugation. For a convention linear in the first argument, formulate and derive the adjoint identity consistently with that convention.

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Check both sides on the same complex vectors



Let $A = \begin{bmatrix} 2i & 3 \\ 1 & -1 \end{bmatrix}$, so $A^H = \begin{bmatrix} -2i & 1 \\ 3 & -1 \end{bmatrix}$. For $x = (1, i)^T$, $y = (2, 1+i)^T$, obtain $Ax = (5i, 1-i)^T$ and $A^H y = (1-3i, 5-i)^T$.

Term by term, $\langle Ax, y \rangle = (-5i)2 + (1+i)(1+i) = -8i$, while $\langle x, A^H y \rangle = (1, -i)(1-3i, 5-i)^T = -8i$. Conjugating the wrong factor can change either result.

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Orthogonality and the four subspaces

Orthogonality means

$$u^H v = 0.$$

For example,

$$u = (1, i)^T, \qquad v = (1, -i)^T,$$
$$u^H v = (1, -i)(1, -i)^T = 1 + (-i)(-i) = 1 - 1 = 0.$$

The complex four-subspace relations use the conjugate transpose.

$$C(A) \perp N(A^H), \qquad C(A^H) \perp N(A).$$

Ordinary A^T does not generally give the Hermitian orthogonal complements. For instance, $(A^H y)^H x = y^H A x$, which vanishes when $Ax = 0$.

Dimensions:

$$\dim C(A) = \dim C(A^H) = r,$$
$$\dim N(A) = n - r, \qquad \dim N(A^H) = m - r.$$

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Hermitian matrices

A square matrix is Hermitian if

$$S^H = S.$$

Entrywise:

$$s_{ji} = \overline{s_{ij}}.$$

Its diagonal entries are real. For every z,

$$(z^H S z)^H = z^H S^H z = z^H S z.$$

The scalar equals its own conjugate, so

$$z^H S z \in \mathbb{R}.$$

For example,

$$S = \begin{bmatrix} 2 & 1+i \\ 1-i & 3 \end{bmatrix}$$

is Hermitian.

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Hermitian eigenvalues and eigenvectors

If nonzero z satisfies

$$Sz = \lambda z,$$

multiply on the left by z^H.

$$z^H S z = \lambda z^H z.$$

The left side is real and $z^H z > 0$, proving λ real.

If

$$Su = \lambda u, \qquad Sv = \mu v,$$

then, using reality of λ,

$$\lambda u^H v = (Su)^H v = u^H S v = \mu u^H v.$$

For distinct eigenvalues,

$$(\lambda - \mu) u^H v = 0,$$

so the eigenvectors are orthogonal.

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The spectral theorem: a complete orthonormal eigenbasis

For completeness of the eigenbasis, maximize the real-valued quadratic form $z^H S z$ on the complex unit sphere. Stationarity in real and imaginary directions gives $Sq = \lambda q$. The complement $q^\perp$ is invariant because $q^H S v = (Sq)^H v = 0$. Restriction remains Hermitian, so induction yields a complete orthonormal eigenbasis. Gram-Schmidt can be applied within repeated eigenspaces. Thus

$$S = Q \Lambda Q^H,$$

where $Q^H Q = I$ and Λ is real diagonal.

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Unitary matrices

A square matrix $Q \in \mathbb{C}^{n \times n}$ is unitary if

$$Q^H Q = I.$$

It also satisfies $QQ^H = I$, so

$$Q^{-1} = Q^H.$$

It preserves inner products.

$$\langle Qu, Qv \rangle = (Qu)^H (Qv) = u^H v.$$

Thus $\|Qu\| = \|u\|$.

If $Qu = \lambda u$,

$$\|u\| = \|Qu\| = |\lambda| \|u\|,$$

so

$$|\lambda| = 1.$$

The determinant also records length preservation.

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Unitary matrices (continued)

$$1 = \det(Q^H Q) = (\det Q)^H \det Q = |\det Q|^2,$$

Hence $|\det Q| = 1$. Real orthogonal matrices are the real special case of unitary matrices.

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Normal matrices

A square matrix is normal if

$$AA^H = A^H A.$$

A complex square matrix is unitarily diagonalizable exactly when it is normal. For sufficiency, take a unit eigenvector and extend it to a unitary basis, producing $\begin{bmatrix} \lambda & b^H \\ 0 & C \end{bmatrix}$. Equality of the $(1,1)$ entries of AA^H and $A^H A$ gives $|\lambda|^2 + \|b\|^2 = |\lambda|^2$, so $b = 0$. The remaining block is normal, and induction proves the result. Necessity follows by multiplying $A = Q\Lambda Q^H$ and its adjoint. Important classes include

- Hermitian:eigenvalues real;
- skew-Hermitian $K^H = -K$:eigenvalues pure imaginary;
- unitary:eigenvalues modulus 1.

Ordinary diagonalizability requires an eigenbasis. Unitary diagonalizability requires an orthonormal eigenbasis, a stronger condition.

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A Hermitian diagonalization example

$$S = \begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix} = S^H.$$

Characteristic polynomial:

$$\det(S - \lambda I) = (2 - \lambda)^2 - |i|^2 = (\lambda - 1)(\lambda - 3).$$

Eigenvalues 1,3.

For $\lambda = 3$,

$$\begin{bmatrix} -1 & i \\ -i & -1 \end{bmatrix} z = 0$$

gives $z_1 = iz_2$. Choose

$$q_1 = (i, 1)^T / \sqrt{2}.$$

For $\lambda = 1$, choose

$$q_2 = (-i, 1)^T / \sqrt{2}.$$

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A Hermitian diagonalization example (continued)

Both vectors have unit norm and $q_1^H q_2 = 0$. Therefore

$$S = Q \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} Q^H,$$

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix}.$$

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The Fourier matrix

Let

$$\omega = e^{2\pi i/n}.$$

Forward Fourier matrix:

$$(F_n)_{jk} = \omega^{jk}, \qquad 0 \le j, k < n.$$

Forward transform:

$$y = F_n c.$$

The Fourier matrix is symmetric. It is generally not Hermitian for $n > 2$; sizes 1 and 2 are real symmetric exceptions. Its columns are orthogonal because

$$\sum_{j=0}^{n-1} (\omega^{jk})^H \omega^{j\ell} = \sum_{j} \omega^{j(\ell-k)} = n\delta_{k\ell}.$$

Therefore

$$F_n F_n^H = nI, \qquad F_n^{-1} = F_n^H/n.$$

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Inverse DFT and Parseval

Inverse:

$$c_j = \frac{1}{n} \sum_{k=0}^{n-1} \omega^{-jk} y_k.$$

Software using

$$\omega = e^{-2\pi i/n},$$

interchanges the forward and inverse sign conventions.

Parseval:

$$\|y\|^2 = (F_n c)^H (F_n c) = n\|c\|^2.$$

Normalized unitary Fourier matrix:

$$Q = F_n/\sqrt{n}$$

satisfies $Q^H Q = I$, preserving the norm. Normalization changes scale constants without changing which information the transform represents.

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DFT and cyclic convolution

Define cyclic convolution by $h_j = \sum_{k=0}^{n-1} a_k b_{(j-k) \bmod n}$. Its Fourier transform is

$$F_n h = (F_n a) \odot (F_n b)$$

Substitute this definition into $\sum_j \omega^{\ell j} h_j$ and set $s = (j-k) \bmod n$. The sum factors into $(\sum_k a_k \omega^{\ell k})(\sum_s b_s \omega^{\ell s})$. This identity uses the unnormalized forward transform. A forward transform divided by $\sqrt{n}$ requires an extra factor $\sqrt{n}$ on the right.

Two FFTs, pointwise multiplication, and an inverse FFT take $O(n \log n)$ arithmetic operations, compared with $O(n^2)$ for general direct cyclic convolution. Sequences of lengths p, q have linear convolution length $p + q - 1$; pad to at least that length to avoid wraparound.

Coefficient order, sign, and normalization must match the convolution definition. Record these conventions in FFT or finite-field transform implementations.

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Even–odd decomposition in radix-2 FFT

Let $n = 2m$, $0 \le j < m$, and split the sequence into

$$c' = (c_0, c_2, \ldots, c_{2m-2}),$$
$$c'' = (c_1, c_3, \ldots, c_{2m-1}).$$

Separate $y_j = \sum_{k=0}^{n-1} c_k \omega_n^{jk}$ into even indices $k = 2s$ and odd indices $k = 2s+1$. Since $\omega_n^2 = \omega_m$, $\omega_n^m = -1$,

$$y_j = y'_j + \omega_n^j y''_j,$$
$$y_{j+m} = y'_j - \omega_n^j y''_j.$$

where y', y'' are DFTs of size $m = n/2$.

This butterfly operation combines two half-sized problems with m twiddle multiplications. One split works for any positive even n; recursively using radix-2 down to size 1 additionally requires $n = 2^\ell$, $\ell \ge 0$.

For $n = 4$, $c = (1, 2, 0, -1)^T$, bit reversal first gives

$$(c_0, c_2, c_1, c_3) = (1, 0, 2, -1).$$

The first stage performs two two-point butterflies.

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Even–odd decomposition in radix-2 FFT (continued)

$$(1,0) \mapsto (1,1), \qquad (2,-1) \mapsto (1,3).$$

The second stage uses $\omega_4^0 = 1, \omega_4^1 = i$.

$$(1,1) \mapsto (2,0), \qquad (1,3) \mapsto (1+3i, 1-3i).$$

Consequently,

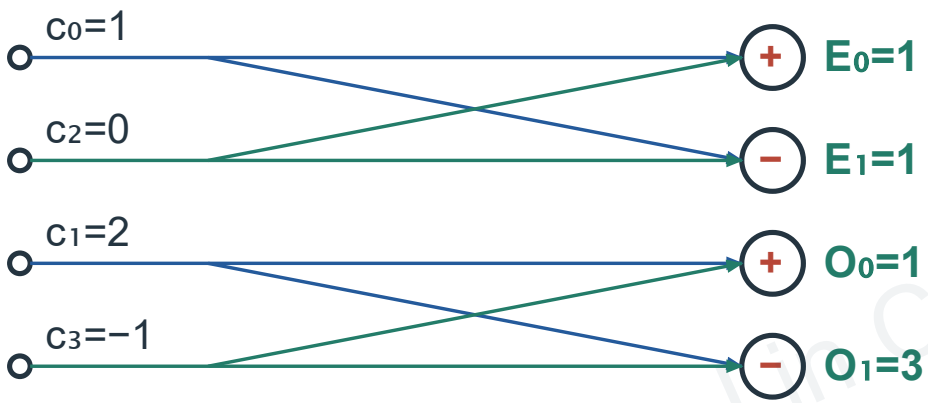
$$y = F_4 c = (2, 1+3i, 0, 1-3i)^T.$$

Direct evaluation gives $y_1 = 1 + 2i + (-1)(-i) = 1 + 3i$, $y_3 = 1 - 2i + (-1)i = 1 - 3i$. The plus and minus branches share the same twiddle product.

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Even–odd decomposition in radix-2 FFT (continued)

Four-point FFT stage 1: even and odd two-point DFTs



Even $(c_0,c_2)=(1,0)$

$E_0=c_0+c_2 \cdot E_1=c_0-c_2$

Odd $(c_1,c_3)=(2,-1)$

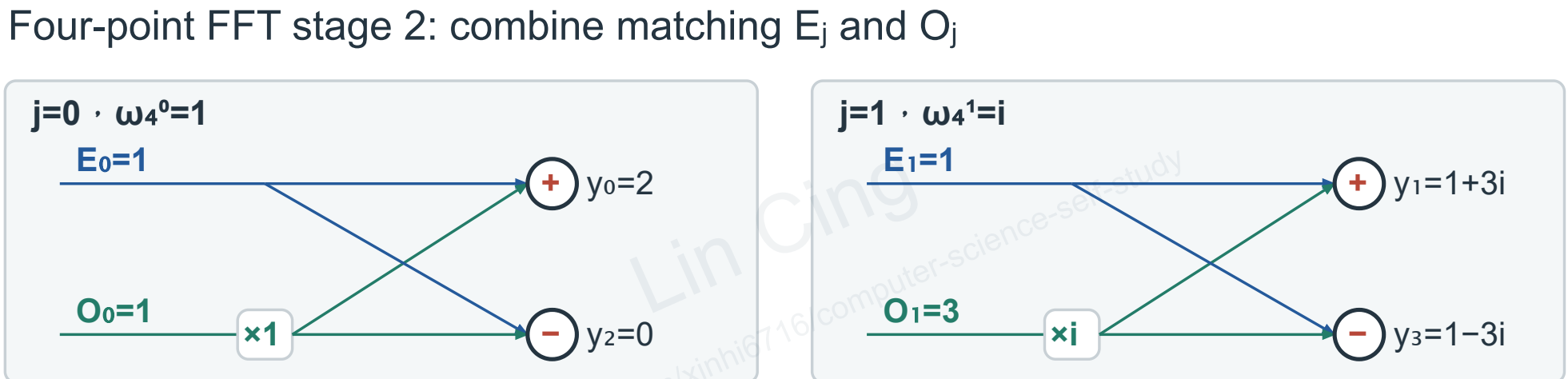
$O_0=c_1+c_3 \cdot O_1=c_1-c_3$

Upper branch adds; lower subtracts green from blue. Next, pair matching E_j and O_j.

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Which intermediate values combine in a four-point FFT?

The first-stage array is $(E_0, E_1, O_0, O_1) = (1, 1, 1, 3)$. The second stage pairs positions $(0, 2)$ and $(1, 3)$. Only matching E_j, O_j produce y_j, y_{j+2}. The diagram separates the two butterflies to keep those pairings visible.



Blue $u = E_j$; green $\omega v = \omega_4^j O_j$. Upper: $u + \omega v$; lower: $u - \omega v$. Output order y_0, y_1, y_2, y_3.

The first three intermediate values happen to agree in this example, concealing a wrong pairing. Instead take $c = (2, 5, 7, -1)$. The first stage is $(9, -5, 4, 6)$, and the correct output is $(13, -5 + 6i, 5, -5 - 6i)$. Pairing E_0, E_1 incorrectly would give $9 + (-5) = 4$ instead of 13. Pairing indices are part of the algorithm.

Recursion and arithmetic cost

$n = 2^\ell$:

$$T(n) = 2T(n/2) + \Theta(n).$$

There are $\log_2 n$ levels with linear work at each level, giving

$$T(n) = \Theta(n \log n).$$

Naive Fourier matrix multiplication:

$$n^2$$

multiplications in the direct matrix formula.

For $n = 1024 = 2^{10}$,

$$n^2 = 1,048,576,$$
$$\frac{n}{2} \log_2 n = 5,120.$$

The theoretical multiplication-count ratio is 204.8 in this model. This is not a measured runtime speedup. Additions, memory access, parallel execution, and root generation affect actual runtime.

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Iterative FFT and bit reversal

The recursion can be implemented bottom-up. A radix-2 decimation-in-time arrangement for $n = 8$ begins with bit-reversed input order

$$0, 4, 2, 6, 1, 5, 3, 7.$$

and applies butterflies of lengths 2, 4, and 8.

$$(a, b) \mapsto (a + \omega b, a - \omega b).$$

An inverse transform can conjugate the input, apply the forward transform, conjugate the output, and divide by n.

Other sizes can use mixed-radix methods or Bluestein's algorithm. A real-input FFT can exploit conjugate symmetry, but a generic complex interface need not do so automatically.

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Numerical interpretation

Practical issues include the following.

- Compute or store twiddle factors accurately.
- Control accumulated phase error.
- Specify the forward scale and inverse sign.
- Zero-padding changes the frequency grid but creates no new observations.
- Windowing changes the signal while reducing some leakage.
- Bin frequency is $k/(n\Delta t)$; bins above Nyquist may instead be represented as $(k-n)/(n\Delta t)$ under a signed-frequency convention. For even n, specify the sign convention for the Nyquist bin.

A spectral peak can combine signal, leakage, and noise. A nonzero numerical bin alone does not establish an exact physical frequency.

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Chapter review

- The first inner-product argument is conjugated; adjoints use the conjugate transpose.
- Hermitian eigenvalues are real; unitary matrices preserve norm, under different conditions.
- State DFT sign and normalization; Parseval scaling depends on the latter.
- Full radix-2 recursion requires a power-of-two length; butterflies pair matching indices.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 9.

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